Lecture 10

Maxwell initial data

The Maxwell equations admit many potentials for the same field. We impose a condition on the potential that turns its equations into wave equations. Two compatibility conditions on the initial data make this condition persist during evolution.

Objectives. *We construct a smooth Maxwell potential from a conserved compact source and compatible compact initial data. We prove propagation of the gauge condition and exhibit the remaining gauge freedom in a vacuum packet.*

By the end of this lecture you will be able to:

- Check the Lorenz and Gauss compatibility conditions on potential data.
- Construct the potential by retarded and homogeneous wave solutions.
- Prove constraint propagation and identify residual gauge waves.

10.1 The Lorenz condition

We retain Lecture 9's potential and fields on $\mathbb{R}^{1+3}$: $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\,\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \mathbf{A}_t$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. The current one-form is $j = \rho\,\mathrm{d}t - \mathbf{J}\cdot\mathrm{d}x$.

Definition 10.1 (Lorenz gauge and constraints)**.** The *Lorenz function* of a potential is

$$\ell = \partial^{\mu}(A_{\mathrm{em}})_{\mu} = -\big(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A}\big). \tag{10.1}$$

The *Lorenz gauge* is the condition $\ell = 0$. A *constraint* is a condition on initial data required by the field equations. *Constraint propagation* means that an initially satisfied constraint remains satisfied under evolution.

The Lorenz function relates the wave operator to the inhomogeneous Maxwell equations. For every C^2 potential,

$$\operatorname{div}\mathbf{E} = \Box\phi_{\mathrm{em}} + \ell_t, \qquad \operatorname{curl}\mathbf{B} - \mathbf{E}_t = \Box\mathbf{A} - \nabla\ell. \tag{10.2}$$

The first equality follows by expanding $\operatorname{div}\mathbf{E}$ and ℓ_t. For the second, use $\operatorname{curl}\operatorname{curl}\mathbf{A} = \nabla\operatorname{div}\mathbf{A} - \Delta\mathbf{A}$. For example, its x component is $\partial_x\partial_y A_y + \partial_x\partial_z A_z - \partial_y^2 A_x - \partial_z^2 A_x$, which is the x component of the stated right side. Cyclic permutation proves the other components. Substitution of the electric field gives (10.2).

Consequently, in Lorenz gauge the reduced potential equations are

$$\Box A_{\mathrm{em}} = -j, \qquad \Box\phi_{\mathrm{em}} = \rho, \qquad \Box\mathbf{A} = \mathbf{J}. \tag{10.3}$$

The wave operator acts on each coefficient of the one-form. Its time coefficient is $-\phi_{\mathrm{em}}$, which explains the scalar source sign.

10.2 Compatible data and their evolution

Write the potential data at time zero as

$$\phi_{\mathrm{em}}(0) = a_\phi, \qquad \partial_t \phi_{\mathrm{em}}(0) = b_\phi, \qquad \mathbf{A}(0) = a_A, \qquad \mathbf{A}_t(0) = b_A.$$

The initial Lorenz condition is $b_\phi + \operatorname{div} a_A = 0$. The initial electric field is $-\nabla a_\phi - b_A$, so Gauss's law requires $-\Delta a_\phi - \operatorname{div} b_A = \rho(0)$.

Theorem 10.2 (Maxwell evolution in Lorenz gauge). *Let ρ and $\mathbf{J}$ be smooth real functions of compact spacetime support, with $\partial_t \rho + \operatorname{div} \mathbf{J} = 0$. Let a_ϕ, b_ϕ, a_A and b_A be smooth compactly supported spatial data, scalar for the first two and vector-valued for the last two. Assume the compatibility conditions*

$$b_\phi + \operatorname{div} a_A = 0, \qquad -\Delta a_\phi - \operatorname{div} b_A = \rho(0, \cdot). \tag{10.4}$$

There is a unique smooth global potential solving (10.3) with these data. It satisfies Lorenz gauge and all Maxwell equations. It is also the unique C^2 Lorenz-gauge Maxwell potential with these potential data.

Proof. We construct the wave solution first and then propagate its Lorenz condition. Set $R = G_{\mathrm{ret}} * (-j)$, taking the convolution coefficient by coefficient. Theorem 6.2 gives a smooth one-form with $\Box R = -j$. Its scalar-potential component is $G_{\mathrm{ret}} * \rho$, and its spatial vector component is $G_{\mathrm{ret}} * \mathbf{J}$.

The data $R(0)$ and $R_t(0)$ are smooth and compactly supported in space. Indeed, enclose the source in $[s_-, s_+] \times \overline{B_L(0)}$. The support bound of Lecture 6 puts $R(t)$ inside $\overline{B_{L+t-s_-}(0)}$ when $t \geq s_-$, and gives zero when $t < s_-$. Each time derivative has the same support bound because it is the retarded convolution with the corresponding derivative of the source.

Let $a = -a_\phi \, \mathrm{d}t + a_A \cdot \mathrm{d}x$ and $b = -b_\phi \, \mathrm{d}t + b_A \cdot \mathrm{d}x$ denote the one-form data. Use the massless construction of Theorem 4.3 to solve

$$\Box H = 0, \qquad H(0) = a - R(0), \qquad H_t(0) = b - R_t(0).$$

The corrected data are smooth and compactly supported, hence Schwartz. The multiplier argument of Lecture 7 gives a smooth spacetime solution for every component. Thus

$$A_{\mathrm{em}} = G_{\mathrm{ret}} * (-j) + H \tag{10.5}$$

solves the reduced wave equations and has exactly the prescribed data. The subtraction is needed whenever the retarded field has nonzero data at time zero. If the source starts later, both corrections vanish.

Next, differentiation commutes with the wave operator on this smooth solution. Current conservation gives

$$\Box \ell = -\partial_t \Box \phi_{\mathrm{em}} - \operatorname{div} \Box \mathbf{A} = -\partial_t \rho - \operatorname{div} \mathbf{J} = 0.$$

The first compatibility condition gives $\ell(0) = 0$. Using $\partial_t^2 \phi_{\mathrm{em}} = \Delta \phi_{\mathrm{em}} + \rho$ in its time derivative gives

$$\ell_t(0) = -\big(\Delta a_\phi + \rho(0, \cdot) + \operatorname{div} b_A\big) = 0$$

by the Gauss condition. Corollary 5.4 now implies $\ell = 0$ on all spacetime. It applies to this smooth scalar wave without a total-energy assumption. Equation (10.2) supplies Gauss's and Ampère's laws. The other two Maxwell equations follow from $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ as in Lecture 9. This proves propagation of the gauge condition and Gauss constraint.

Finally, two C^2 reduced wave solutions with the same data have a homogeneous difference with zero data. The same local uniqueness corollary makes every component of the difference zero. Every C^2 Lorenz-gauge Maxwell potential obeys the reduced wave equations by (10.2), so this also proves the final uniqueness assertion. ∎

The compact-current assumption restricts total charge. Define $Q(t) = \int_{\mathbb{R}^3} \rho(t,x)\,\mathrm{d}^3x$. The current has uniformly bounded spatial support, so differentiation and coordinate integration by parts give

$$Q'(t) = -\int_{\mathbb{R}^3} \operatorname{div}\mathbf{J}(t,x)\,\mathrm{d}^3x = 0.$$

The source vanishes sufficiently early, hence $Q(t) = 0$ for every t. The theorem does not cover a source with nonzero total charge. A Coulomb potential with tail $Q/(4\pi|x|)$ for $Q \neq 0$ is not in L^2, hence not in H^1 at spatial infinity: its squared radial integral contains a nonzero constant integrated over $1 < r < \infty$. Sheet 10 treats that different data class.

10.3 A transverse packet and a residual gauge wave

Example 10.3 (A compact vacuum packet)**.** Take $\rho = 0$ and $\mathbf{J} = 0$. Choose a smooth compactly supported cutoff ϑ equal to one near the origin, and put $f(x,y,z) = (1 - x^2 - y^2)\vartheta(x,y,z)$. Let u be the massless scalar solution of Lecture 4 with $u(0) = f$ and $u_t(0) = 0$. Define

$$\phi_{\mathrm{em}} = 0, \qquad \mathbf{A} = \operatorname{curl}(ue_z) = (u_y, -u_x, 0), \qquad e_z = (0,0,1).$$

The initial vector potential is $(f_y, -f_x, 0)$ and its initial velocity is zero. Its divergence vanishes by commutation of mixed derivatives. Both compatibility conditions therefore hold. Differentiation of $\Box u = 0$ shows that every component of $\mathbf{A}$ solves the wave equation. The potential is in Lorenz gauge because $\phi_{\mathrm{em}} = 0$ and $\operatorname{div}\mathbf{A} = 0$ for every time. Its fields are

$$\mathbf{E} = (-u_{ty}, u_{tx}, 0), \qquad \mathbf{B} = (u_{xz}, u_{yz}, -u_{xx} - u_{yy}).$$

At time zero, $\mathbf{E} = 0$ and $B_z(0,0) = 4$. The packet is nonzero. If $\operatorname{supp} f \subset \overline{B_L(0)}$, finite propagation in Lecture 5 puts these fields inside $\overline{B_{L+|t|}(0)}$ at each time.

A gauge transformation preserves Lorenz gauge exactly when its function satisfies a wave equation. Direct substitution into (10.1) gives

$$\ell \longmapsto \ell + \Box\chi. \tag{10.6}$$

A *residual gauge wave* is a smooth solution of $\Box\chi = 0$. For the packet, choose such a wave with $\chi(0) = 0$ and $\chi_t(0) = p \in C_c^\infty(\mathbb{R}^3)$, where $p \neq 0$. The transformed potential has data

$$a'_\phi = -p, \qquad b'_\phi = 0, \qquad a'_A = (f_y, -f_x, 0), \qquad b'_A = \nabla p.$$

Its Lorenz condition is unchanged, and its Gauss condition reads $-\Delta(-p) - \operatorname{div}\nabla p = 0$. The transformed fields are the same because $\mathrm{d}^2\chi = 0$. Residual gauge waves change potential data while preserving electric and magnetic fields.

10.4 Spatial transversality alone is insufficient

The condition $\operatorname{div} \mathbf{A} = 0$ is often called spatial transversality. It does not impose the scalar condition or Gauss's law. In vacuum, take $a_A = b_A = 0$ and $a_\phi = b_\phi = h$, where $h \in C_c^\infty(\mathbb{R}^3)$ is nonzero and $\Delta h \neq 0$. The reduced wave solution has $\mathbf{A} = 0$ for every time, hence is spatially transverse. At time zero, however,

$$\ell(0) = -h \neq 0, \qquad \operatorname{div} \mathbf{E}(0) = -\Delta h \neq 0.$$

Both required conditions fail. The scalar potential and vector velocity must enter the compatibility test. Lectures 11 and 12 distinguish these conditions in the Hamiltonian formulation.

Summary. A smooth conserved current of compact spacetime support and compatible smooth compact potential data determine a unique Lorenz-gauge Maxwell potential. The construction adds a retarded source field to a homogeneous wave with corrected initial data. The Lorenz function obeys the homogeneous wave equation. Its zero displacement and velocity are the Lorenz and Gauss compatibility conditions. Residual gauge waves preserve the fields while changing potential data. Spatial transversality alone supplies neither condition.

Self-check. For data $(a_\phi, b_\phi, a_A, b_A)$, compute $\ell(0)$ and $\ell_t(0)$ using the reduced wave equation. Which condition makes each one zero?

Answer. The values are $-(b_\phi + \operatorname{div} a_A)$ and $-(\Delta a_\phi + \rho(0) + \operatorname{div} b_A)$. The initial Lorenz condition kills the first. Initial Gauss compatibility kills the second.

Self-check. The retarded source field may have nonzero data at time zero. What goes wrong if the homogeneous field is assigned the original data a and b without correction?

Answer. The sum then has data $a + R(0)$ and $b + R_t(0)$. Its wave equation is correct, but its data are wrong. Subtracting both retarded data gives the required a and b.

Self-check. Why does compact spacetime support, together with conservation, force zero total charge? Does conservation alone force it?

Answer. Conservation makes the total charge constant, and compact time support makes it zero sufficiently early. Conservation alone allows a nonzero constant charge when the spatial integral and vanishing boundary flux are defined.

Self-check. A smooth gauge function satisfies $\Box\chi \neq 0$. Does it preserve the electric and magnetic fields? Does it preserve Lorenz gauge?

Answer. It preserves both fields because $\mathrm{d}^2\chi = 0$. It changes the Lorenz function by $\Box\chi$, so it does not preserve Lorenz gauge. A residual gauge wave has $\Box\chi = 0$.